

STUDENT LEARNING COMPANION

Quantitative Methods for Business Analysis

Orientation, Readiness, and Question Framing

GC1015 | S01

Primer + PNP-IND + PNP-GRP + Section Notes



Arrive with an initial point of view,
then revise it as the evidence
changes.

Primer and Participation Guide

Prepare before class, retrieve during class, and reflect after each section

- Complete the four-question Primer before class.
- Use Guided Recall pages after concept clusters, not as transcription pages.
- Use each Section Notes page for takeaways, clarity needs, and applications.
- Complete PNP-GRP with your team and keep assumptions visible.
- Finish with the individual Exit Ticket.



The goal is retrieval and prediction,
not perfection.

Primer Check

Commit to an answer before class discussion

1

True or False: According to the primer, opening analytical software like Excel or SPSS before defining the decision owner, pending action, and evidence boundary is an effective way to quickly discover the right business question.

Your response

Capture your initial thinking and brief rationale here.



The goal is retrieval and prediction,
not perfection.

Primer Check

Commit to an answer before class discussion

2

True or False: OmniRetail Customer Operations proposed spending \$450,000 annually to hire 40 weekend customer support representatives to address a Monday morning backlog of 4,200 unresolved tickets.

Your response

Capture your initial thinking and brief rationale here.



The goal is retrieval and prediction,
not perfection.

Primer Check

Commit to an answer before class discussion

3

True or False: A Monday morning queue of 4,200 unresolved tickets empirically proves that the tickets were generated by customers submitting support requests over the weekend.

Your response

Capture your initial thinking and brief rationale here.



The goal is retrieval and prediction,
not perfection.

Primer Check

Commit to an answer before class discussion

4

True or False: In the OmniRetail case, a statistically significant difference of 0.3 percentage points in cancellation rates would justify committing the \$450,000 operational budget because statistical significance guarantees material financial return.

Your response

Capture your initial thinking and brief rationale here.



GC1015

QUANTITATIVE METHODS FOR BUSINESS ANALYSIS

Orientation, Readiness, and Question Framing

WELCOME TO THE COURSE

From business questions to evidence-based decisions



LEARNING OBJECTIVES

Frame business questions, choose sound statistical methods, and communicate defensible findings.



LEARNING APPROACH

Hands-on analysis, short cases, tool demonstrations, and structured discussion.



TOOLS

Excel and IBM SPSS, with Python used for selected demonstrations.



MINDSET

Question, measure, analyze, decide.



THE QUESTION

Begin with a decision worth making.



THE METHOD

Choose an approach that fits the question.



THE EVIDENCE

Analyze and validate what the data supports.



THE DECISION

Communicate a defensible recommendation.

ABOUT YOUR PROFESSOR

Academic and professional perspective



ACADEMIC HISTORY

- PhD in Transportation and Supply Chain, with a modeling and data focus
- MBA in Business Analytics
- MS in Construction Management
- certificates in Business Analytics and Statistics.



INDUSTRY EXPERIENCE

- Analytics management consulting
- 6+ years in transportation planning and analysis
- 8+ years in data analysis
- 5+ years in project management.

ABOUT YOU

A brief introduction to the learning community

- 1 What name would you like us to use?
- 2 Where are you joining us from?
- 3 What is your academic or professional background?
- 4 What do you hope to gain from this course?



**KEEP IT TO ABOUT ONE
MINUTE**

BUILD YOUR PROFESSIONAL STORY

Connect course work to résumé-ready evidence

YOUR RÉSUMÉ-READY CAPABILITY

Turn raw business data into recommendations using statistical reasoning, spreadsheet analysis, SPSS, and clear visual communication.

ROLE CONNECTIONS

Data Analyst

BI Analyst

Strategy Analyst

Operations Analyst



KEEP THE EVIDENCE

completed analyses, model decisions, validation checks, concise findings, and reflections on tradeoffs.

WHAT IS GC1015?

Start with your current mental model, then unpack the title

What comes to mind when you hear GC1015?

WHAT WORDS, EXAMPLES, OR EXPERIENCES COME TO MIND?

1

QUANTITATIVE

measurable evidence

2

METHODS

structured ways to analyze

3

BUSINESS

a real decision or objective

4

ANALYSIS

interpreting what the
evidence means

A WORKING DEFINITION

Connect this course to systems and decisions you already...

WORKING DEFINITION

Quantitative methods for business analysis use numerical evidence and structured analytical techniques to understand performance, evaluate uncertainty, and support decisions.

YOU MAY HAVE ALREADY ENCOUNTERED THIS TODAY



Your phone summarizes
screen time



A map estimates your
arrival time



A retailer tests a promotion



A fitness app compares
your trends over time



Three Pillars of Decision-Grade
Analysis

SESSION ROADMAP

Three Pillars of Decision-Grade Analysis

1

Section 01: Decision Discipline 🎯

- Distinguish business problems, stakeholder decisions, analytical questions, and software tasks.
- Establish the eight-step analytical lifecycle before touching data.

2

Section 02: Bounded Questions 📐

- Formulate falsifiable claims and define rigorous evidence boundaries.
- Audit assumptions, disclose limitations, and avoid causal overreach.

3

Section 03: Tool Readiness 🖥️

- Verify Minimum Viable Readiness (**MVR**) in Excel and **SPSS**.
- Complete guided practice, group studio, and synthesis after the break.



Move from Computation to
Decision Value

THE ANALYST'S MANDATE

Move from Computation to Decision Value

1

The Data Paradox

- More data does not automatically produce better managerial decisions.
- Irrelevant data introduces cognitive noise and increases organizational risk.

2

The Tool-First Trap

- Opening software before framing the choice answers the wrong question.
- High-speed computation cannot rescue an unanchored analytical request.

3

The Professional Mandate

- Evidence must reduce uncertainty for an accountable decision maker.
- Align metrics, grain, and methods directly with executive choices.

SECTION

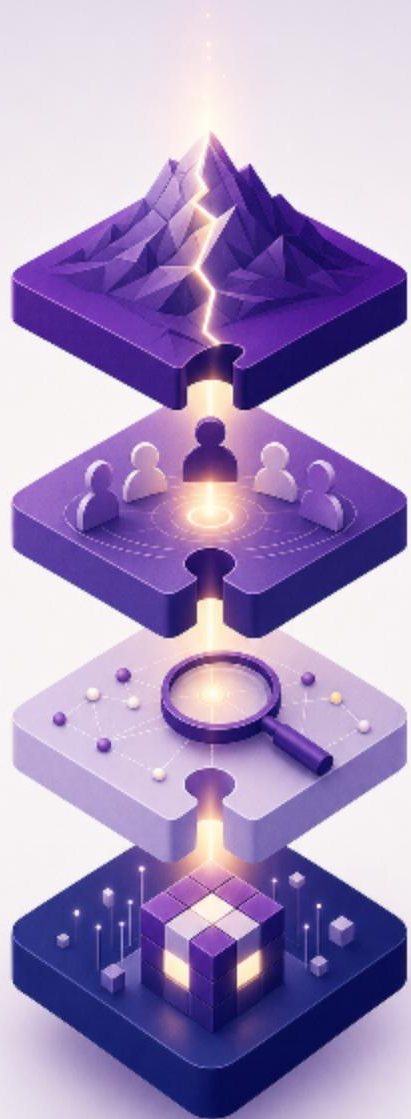
SECTION 01: DECISION DISCIPLINE

Separate Problems, Decisions, Questions, and Tasks



SESSION CASE: OMNIRETAIL

A Simulated Customer Retention Decision



FOUR LEVELS OF AN ANALYTICAL REQUEST

Problem, Decision, Question, and Task

1

Problem vs. Decision 🎯

- **Business Problem** An observed condition requiring attention, such as increasing customer churn.
- **Stakeholder Decision** A resource commitment, such as funding \$450,000 for weekend staff.

2

Question vs. Task 💻

- **Analytical Question** The evidence needed, such as testing weekend ticket cancellation links.
- **Software Task** The mechanical execution, such as summarizing records in statistical software.

3

Strategic Alignment 🔗

- Aligning every computation with executive decision rights eliminates wasted modeling effort.
- Mechanical execution without decision alignment creates zero organizational value.



The Spine of Quantitative
Analysis

THE EIGHT-STEP WORKFLOW

The Spine of Quantitative Analysis

1

Frame and Inspect 🔍

- **Frame** Clarify the business problem, stakeholder decision, and metric definition.
- **Inspect** Audit data provenance, observation grain, missingness, and baseline quality.

2

Choose, Assume, and Analyze ⚙️

- **Choose & Assume** Select valid methods and document required operational conditions.
- **Analyze** Execute transparent, reproducible calculations without buried manual steps.

3

Bound and Communicate 📢

- **Uncertainty & Recommend** Quantify decision uncertainty and derive bounded action items.
- **Communicate** Translate empirical findings into defensible executive recommendations.



EVIDENCE AND MANAGERIAL JUDGMENT

Distinct Roles in an Accountable Decision

1

Quantitative Evidence 📊

- **Empirical Patterns** Statistical distributions, point estimates, effect sizes, and uncertainty bounds.
- **Analytical Role** Informs decision makers by systematically reducing uncertainty around key variables.

2

Managerial Judgment ⚖️

- **Strategic Context** Organizational feasibility, risk tolerance, legal constraints, and ethics.
- **Executive Role** Integrates empirical findings with budget limits and operational capabilities.

3

The Decision Boundary 🎯

- **Clear Separation** Data never makes a decision; accountable human leaders choose actions.
- **Professional Standard** Statistical significance does not equal mandatory operational action.

REFRAMING A TOOL-FIRST REQUEST

Move from Keystrokes to Strategic Intent

1

The Flawed Request ❌

- **Raw Command** "Run **SPSS** on every support ticket and stop customer churn."
- **Defects** Lacks decision ownership, scope boundaries, baseline metrics, and thresholds.

2

The Diagnostic Repair 🔍

- **Identify Choice** Anchor analysis to the pending \$450,000 weekend staffing decision.
- **Define Comparison** Compare 60-day cancellation rates between weekend and weekday tickets.

3

The Decision-Grade Reframe ✅

- **Structured Question** For monthly subscribers, do unresolved weekend tickets increase churn?
- **Actionable Output** Provides an auditable empirical test to guide leadership investment.



SCENARIO: THE BACKLOG DILEMMA

What Evidence Should OmniRetail Inspect?

SECTION

SECTION 01 SYNTHESIS

Retrieve, Apply, and Transition

SECTION

SECTION 02: BOUNDED QUESTIONS

Define Claims and Evidence Boundaries

ANATOMY OF A BOUNDED QUESTION

Eight Components that Control Scope

1

Decision Context

- **Decision & Threshold** Identify the accountable leader, pending choice, and minimum actionable difference.
- **Organizational Anchor** Anchor inquiry to concrete capital, staffing, or policy commitments.

2

Evidence Scope

- **Population & Period** Specify exact observation units, geographical boundaries, and time horizons.
- **Exclusion Criteria** Document excluded categories explicitly to prevent unmanaged scope expansion.

3

Measurement Plan

- **Measures & Sources** Define observable outcome metrics, comparison baselines, and verified data sources.
- **Audit Trail** Confirm data provenance and pipeline feasibility before starting data manipulation.



A testable claim is a precise, unambiguous statement regarding observable empirical patterns.

TESTABLE CLAIMS AND FALSIFIABILITY

Define What Would Change Your Mind

1

Empirical Testability 🎯

- **Observable Assertions** Ground provisional claims in measurable operational variables and distinct comparison baselines.
- **Concrete Definitions** Replace subjective terms like satisfaction with concrete operational metrics.

2

Falsifiability Principle ⚖️

- **Disconfirmation Criteria** Precommit to specific empirical outcomes that would disprove the provisional hypothesis.
- **Cognitive Protection** Guard against confirmation bias by defining failure thresholds prior to data inspection.

3

OmniRetail Operational Claim 🏢

- **Structured Hypothesis** Weekend-ticket subscribers exceed weekday churn by at least five percentage points.
- **Actionable Rule** An observed difference below five points disconfirms the rationale for weekend staffing.



ASSUMPTIONS AND LIMITATIONS

Different Risks Require Different Responses

1

Inspectable Assumptions 🔍

- **Provisional Premises** Operational conditions accepted provisionally that can and must be audited.
- **Verification Protocol** Validate timestamp integrity, data grain consistency, and source accuracy.

2

Scope Limitations ⚠️

- **Structural Boundaries** Unavoidable constraints that restrict the generalizability of conclusions.
- **Disclosure Protocol** Document non-transferability across enterprise clients or seasonal periods.

3

Governance Architecture 🛡️

- **Active Management** Test and stress-test assumptions; transparently disclose all limitations.
- **Executive Transparency** Prevent decision makers from applying findings beyond valid evidence bounds.

CAUSAL CLAIMS AND OBSERVATIONAL DATA

Use Language that Matches the Design

1

Observational Realities 📊

- Historical logs reveal empirical associations and sequences, not direct causal levers.
- Confounding factors and selection biases frequently distort observed customer behavior.

2

The Causal Trap ⚠️

- Managers naturally seek actionable root causes to justify costly operational investments.
- Asserting that service friction caused cancellations overstates observational evidence.

3

Disciplined Reporting 🛡️

- Frame conclusions with precise language such as "was associated with" or "differed across groups."
- Transparently document alternative explanations to protect organizational decision quality.



DEEPEN THE IDEA



Price Sensitivity in Subscription
Data

HOW CONFOUNDING CREATES A FALSE STORY

Price Sensitivity in Subscription Data

1

The Surface Pattern 🚪

- Subscribers redeeming promotional discount codes exhibit 35% higher annual churn.
- Intuitive management takeaway suggests eliminating discounts to protect subscriber retention.

2

The Confounding Driver 🧠

- Underlying customer price sensitivity influences both coupon redemption and churn.
- Price-sensitive shoppers actively seek discounts and cancel immediately when rates normalize.

3

Governance Guardrail 🛡️

- Eliminating discounts will not make price-sensitive consumers loyal long-term subscribers.
- List at least two credible alternative explanations before presenting observational findings.

THREE SCOPING MISTAKES

Errors that Undermine Analytical Credibility

1

Uncontrolled Scope Drift 🧭

- Stakeholders insert new questions, cohorts, and metrics without adjusting timelines.
- Expanding analysis beyond the initial charter dilutes rigor and delays decisions.

2

Premature Metric Coding ⚡

- Building formulas before securing alignment on operational definitions creates rework.
- Unreconciled definitions produce conflicting reports across cross-functional teams.

3

Concealed Limitations 🙊

- Omitting known data gaps or sample exclusions erodes executive confidence.
- Leaders make high-stakes commitments assuming findings apply universally.



CHECKPOINT: AUDIT THE CLAIM

Which Evaluation Is Professionally Defensible?

1

The Executive Claim 📋

- "Hiring 20 weekend agents will eliminate the 14% increase in cancellations."
- Review the phrasing to evaluate its methodological and professional defensibility.

2

Evaluative Options ⚖️

- **A** Defensible because it directly proposes a concrete managerial action.
- **B** Flawed because it asserts absolute causality and guarantees outcomes.

3

Decision Verdict 🎯

- **Option B is Correct** Absolute guarantees overstep observational evidence.
- Propose a bounded operational pilot rather than promising total churn elimination.



SCENARIO: A BIASED WORKFORCE REQUEST

Frame the Comparison Before Seeing Results

SECTION

SECTION 02 SYNTHESIS

Audit Boundaries Before Analysis

SECTION

SECTION 03: TOOL READINESS

Produce Inspectable Evidence of Execution



EXCEL AND SPSS

Complementary Roles in Business Analysis

1

Excel Modeling Environment

- Build flexible business models, financial projections, dynamic scenarios, and visual summaries.
- Test operational assumptions interactively across interconnected worksheet structures.

2

SPSS Statistical Environment

- Enforce structured variable metadata, measurement levels, and formal statistical hypothesis testing.
- Execute reproducible analytical workflows through saved command syntax and detailed log files.

3

Dual-Tool Integration

- Preserve pristine raw source files and transfer clean datasets across platforms seamlessly.
- Deploy each software application for its strongest methodological and governance capability.



Separate Inputs, Logic, and
Outputs

SPREADSHEET MODELING STANDARDS

Separate Inputs, Logic, and Outputs

1

Dedicated Input Layer

- Isolate all business assumptions, unit costs, and parameters in dedicated labeled cells.
- Never hardcode operational values inside downstream computational formulas or functions.

2

Transparent Logic Architecture

- Construct dynamic formulas referencing isolated inputs and descriptive named ranges exclusively.
- Eliminate buried mathematical constants to ensure auditable, inspectable calculations.

3

Auditable Output Summary

- Format primary decision metrics and Key Performance Indicators (KPIs) in prominent summary blocks.
- Execute reversible recalculation tests to verify that model outputs update dynamically.

SPSS WORKSPACE ARCHITECTURE

Data, Syntax, and Output Are Separate

1

Data Editor Workspace 📁

- Manage case records in Data View and configure variable metadata within Variable View.
- Define precise measurement levels, variable labels, value codings, and missing-value flags.

2

Syntax Editor Engine 💻

- Write, edit, save, and batch-execute reproducible statistical command syntax files.
- Use the Paste button within procedure dialog boxes to generate auditable command code.

3

Output Viewer Log 📄

- Review generated statistical tables, diagnostic charts, execution logs, and system warnings.
- Save and export independent output files without altering underlying data or syntax.





MVR QUALITY CHECKLIST

Verify Source, Execution, and Evidence

1

Source Protection

- Preserve raw source files in read-only storage; execute transformations exclusively on working copies.
- Apply concrete file naming conventions such as PNP01_JaneDoe_jd1234_ReadinessLog.xlsx without placeholder braces.

2

Dynamic Execution

- Execute reversible recalculation tests in Microsoft Excel to verify dynamic formula dependencies.
- Run saved **IBM SPSS** Statistics command syntax to confirm statistical procedures execute without errors.

3

Evidence Verification

- Reopen saved workbooks, syntax scripts, and output logs to confirm cross-session file persistence.
- Export finalized summary tables to verify inspectability and formatting before analytical submission.



Execute, Save, Reopen, and
Inspect

DEMONSTRATION: READINESS TRACE

Execute, Save, Reopen, and Inspect

1

Excel Dynamic Trace

- Open working copy, modify a designated input parameter, and verify linked formula updates.
- Restore original baseline input, verify dependent metrics return to baseline, and save workbook.

2

SPSS Syntax Trace

- Open dataset and syntax script, execute command block, and inspect Output Viewer logs.
- Verify statistical frequency tables generate without warnings, then save .spv output.

3

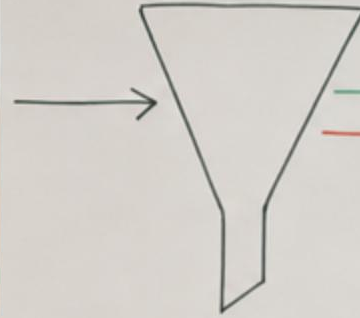
Audit Logging

- Record confirmed file paths, execution timestamps, and software versions in your readiness log.
- Document any configuration errors or missing dependencies immediately for diagnostic resolution.

RAW INGEST

0012345
0009876
1234567
0123456
0001234
9876543
0000001
5556667

VALIDATION FILTER



VALID (TEXT)

0012345
0009876
0123456
0001234
0000001

INVALID (TRUNCATED)

1234567
9876543
5556667

SCENARIO: MISSING LEADING ZEROS

Protect Identifier Integrity During Import

SECTION

SECTION 03 SYNTHESIS

Prove that the Workflow Is Ready



Pause. Reset. Return ready.



SCHEDULED CLASS BREAK

Refresh, Reset, and Confirm File Access

Stretch, Hydrate, and Reset



PRACTICE

Guided Application in Excel and SPSS



GROUP STUDIO

Rotating Decision and Quality Roles

10-Minute Group Work (Session 01)

When a Small Improvement May Be Random Noise

- Northstar piloted a coaching program and observed a higher average customer-satisfaction score for the pilot team. Service representatives, customers, and operations managers will be affected by the expansion decision. The pilot and comparison groups differ in size, their score variability is not the same, and managers want a recommendation this week. No formal test or practical-effect threshold has yet been agreed upon. Your group must decide what quantitative analysis and decision rule should be required before expanding the program, while leaving the final interpretation open for discussion.



Read first. Decide what matters
before proposing a solution.



Share the work. Challenge the reasoning. Align on one recommendation.

Group Decision Tasks

Use evidence, make assumptions visible, and choose a defensible action

- What decision must the organization make, and what evidence matters most?
- What assumption or missing context could change the recommendation?
- Which of two defensible options best fits the stakeholder need, and why?
- What is the largest risk, and which two checks support confidence?



Prepare a concise explanation that another team can audit.

Group Response Workspace

Capture the decision, evidence, tradeoffs, and final recommendation

- Decision: What should the organization do?
- Evidence: Which facts support the decision?
- Tradeoff: What risk or limitation remains?

Group workspace

Capture the group decision, evidence, and reasoning.

WEEK 01 SYNTHESIS

Three Standards to Carry Forward

- **Decision Primacy**

Anchor every quantitative investigation to an accountable executive and a pending choice. Formulate measurable analytical questions before extracting data or selecting statistical tools.

- **Boundary Discipline**

Establish explicit population scopes, exclusions, and precommitted disconfirmation criteria. Manage assumptions in an audit log and match reporting language strictly to research design.

- **Technical Governance**

Preserve pristine raw source files and execute all modifications inside working copies. Validate spreadsheet logic dynamically and preserve reproducible command syntax in SPSS.



Looking Ahead to Week 02

THANK YOU

Looking Ahead to Week 02

1

Next Session Focus

- Examine Key Performance Indicators (KPIs), operational definitions, and observation grain.
- Conduct visual chart critiques and evaluate the statistical integrity of business dashboards.

2

Preparation Protocol

- Complete assigned readings and review the Week 02 starter dataset on Brightspace.
- Verify local installation of Microsoft Excel and **IBM SPSS** Statistics before next class.

3

Academic Support

- Attend scheduled weekly office hours or post technical questions on the Brightspace forum.
- Reference Appendix A, Appendix B, and Appendix C for independent review and extension.



Sources for Concepts, Tools,
and Examples

COURSE AND TECHNICAL REFERENCES

Sources for Concepts, Tools, and Examples

1

Academic Foundation 📖

- Albright and Winston for spreadsheet modeling; Andy Field for applied statistics.
- Cole Nussbaumer Knaflic for data storytelling and visual communication principles.

2

Official Tool Documentation 🖥️

- Microsoft Excel support documentation for Power Query and data type integrity.
- **IBM SPSS** Statistics Core System User Guide for syntax commands and workspace architecture.

3

Industry Evidence Sources 🏭

- Public corporate filings and disclosures from Walmart, The Home Depot, and Delta Air Lines.
- Official corporate newsrooms and investor relations operational reports.

APPENDIX A: THE FOUR LEVELS

Plain-Language Review and Micro-Example

1

Conceptual Framework 🎯

- **Problem vs. Decision** Problem is the observed pain; decision is the resource commitment.
- **Question vs. Task** Question defines needed evidence; task executes software commands.

2

Delivery Micro-Example 🚚

- **Problem & Decision** Delivery delays increase; leadership decides whether to lease additional vans.
- **Question & Task** Question tests route bottlenecks; task calculates dispatch delay averages.

3

Foundation Self-Check ?

- Which of the four levels commits organizational budget, capital, or operational policy?
- The stakeholder decision commits resources; calculations and tasks never commit capital independently.

APPENDIX A: BOUNDED QUESTIONS

Plain-Language Scope Control

1

Core Anatomy

- **Decision & Population** State the accountable leader, choice, and target subscriber cohort.
- **Exclusions & Horizon** Exclude irrelevant subgroups and define explicit observation windows.

2

OmniRetail Micro-Example

- **Framed Question** Compare 60-day cancellation rates between weekend and weekday tickets.
- **Missing Boundaries** Initial drafts often omit subscriber tenure tiers and calendar boundaries.

3

Foundation Self-Check ?

- What critical risk arises when an analyst omits the observation time horizon?
- Without time horizons, seasonal volatility and cohort aging distort cancellation metrics.



What Can Be Checked and
What Must Be Disclosed

APPENDIX A: ASSUMPTIONS AND LIMITS

What Can Be Checked and What Must Be Disclosed

1

Inspectable Assumptions 🔍

- Document provisional premises that can and must be audited against data.
- **Example:** verify that customer ticket timestamps accurately capture initial issue submission.

2

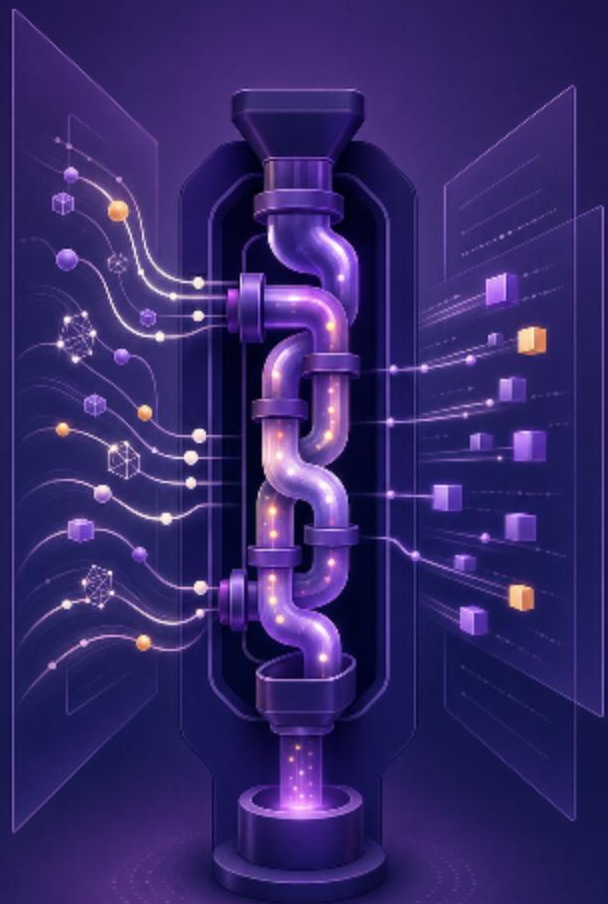
Structural Limitations ⚠️

- Identify fixed project boundaries that restrict the generalizability of findings.
- **Example:** disclose that monthly consumer subscriber models do not apply to enterprise clients.

3

Governance Self-Check ?

- Which analytical condition belongs in the formal project assumption log?
- Timestamp accuracy is an assumption to test; client scope is a limitation to disclose.



APPENDIX A: MINIMUM VIABLE READINESS

Evidence that the Tools Actually Work

1

Observable Execution

- Prove computational readiness through dynamic formula recalculation and executed syntax scripts.
- Move beyond passive software launching by generating verifiable, inspectable analytical artifacts.

2

Artifact Persistence

- Verify that saved workbooks, syntax files, and output logs reopen without corruption.
- Maintain pristine raw sources and conduct transformations exclusively inside working copies.

3

Readiness Self-Check ?

- Why is submitting a static software screenshot insufficient evidence of readiness?
- Screenshots fail to prove that underlying formulas recalculate dynamically or syntax executes cleanly.



APPENDIX B: SCOPE NEGOTIATION

When Stakeholders Want Everything

1

The Expansion Pressure

- Business stakeholders frequently request added variables, customer cohorts, and historical periods.
- Every added analytical dimension multiplies data collection costs, timeline delays, and modeling risk.

2

Scope Trade-Off Matrix

- **Deliver Now** Core bounded question informing the immediate high-stakes decision.
- **Phase Later** Secondary exploratory questions deferred to subsequent analytical cycles.

3

Professional Negotiation

- Frame scope boundaries around stakeholder decision timelines rather than arbitrary resistance.
- Quantify the trade-off in delivery time, data requirements, and analytical clarity.



APPENDIX B: PRACTICAL THRESHOLDS

Statistical Difference Is Not Decision Value

1

Detectability vs. Value

- Large datasets make trivial empirical differences statistically detectable at low p-values.
- Practical significance evaluates whether an observed effect size justifies operational capital expenditure.

2

OmniRetail Case Reality

- Weekend cancellations are 0.3 percentage points higher, achieving statistical significance at scale.
- A 0.3-point reduction preserves \$24,000 annually, failing to justify a \$450,000 budget.

3

Precommitted Decision Rules

- Establish the minimum actionable difference with stakeholders prior to querying production databases.
- Precommitment prevents post-hoc rationalization of economically negligible findings to rescue pet initiatives.



A Responsible Next Step for
Causal Questions

APPENDIX B: FROM ASSOCIATION TO PILOT

A Responsible Next Step for Causal Questions

1

The Observational Dilemma

- Observational records reveal empirical correlations but cannot guarantee direct causal mechanisms.
- Committing \$450,000 immediately or doing nothing both introduce substantial organizational risk.

2

Controlled Operational Pilot

- Implement weekend customer support across three representative regional fulfillment hubs for 90 days.
- Measure First-Contact Resolution (**FCR**), repeat contacts, and 60-day cancellation rates against control hubs.

3

Precommitted Guardrails

- Lock pilot duration, staffing rules, and quantitative stopping criteria before launching operations.
- Prevent mid-experiment rule changes that undermine experimental validity and credibility.



APPENDIX B: CROSS-TOOL REPRODUCIBILITY

Match Evidence Before Adding Complexity

1

Multi-Tool Consistency

- Independent computational tools must produce identical summary statistics from identical data.
- Software discrepancies typically stem from hidden filtering rules or conflicting missing-value treatments.

2

Reconciliation Protocol

- Verify raw record counts, column data types, and filtering logic before comparing descriptive metrics.
- Align aggregation formulas and variance definitions across Microsoft Excel and **IBM SPSS** Statistics.

3

Toolchain Governance

- Microsoft Excel and **IBM SPSS** Statistics form the mandatory, verified core analytical toolchain.
- Optional Python and R scripts serve as reference checks without replacing required deliverables.



APPENDIX C: WALMART SUPPLY CHAIN

Bound Automation Outcomes to Specific Operations

1

Documented Context 🏢

- Walmart announced five high-tech United States perishable distribution centers in 2024.
- Automation modernized grocery distribution across key regional retail fulfillment networks.

2

Reported Performance 📦

- High-tech centers store twice the cases and process more than double traditional volume.
- Physical facility capacity increased significantly without expanding physical building footprints.

3

Analytical Bounding 📐

- Separate perishable facility evidence from broader corporate enterprise capital decisions.
- Define precise baseline metrics, facility types, and observation periods before generalizing.



APPENDIX C: THE HOME DEPOT

Connect Route Decisions to Measurable Results

1

Documented Initiative

- Project Sync optimized transportation routing across United States store replenishment networks.
- Dedicated logistics technology synchronized vendor shipments with regional distribution hubs.

2

Reported Mileage Savings

- The Home Depot reported eliminating 4,000 trucks and 18 million transit miles.
- Operational route consolidation reduced diesel consumption and highway congestion.

3

Empirical Attribution

- Audit baseline periods, route definitions, and freight volume before claiming causation.
- Disclose potential confounders including broader network redesigns and demand fluctuations.



APPENDIX C: DELTA OPERATIONS

Use Multiple Metrics to Bound Performance

1

Documented Performance ✈️

- Delta reported strong operational reliability throughout the June quarter of 2024.
- The carrier recorded 39 cancel-free, brand-perfect operational days across mainline routes.

2

Multi-Metric Measurement 📊

- Evaluate completion factor, on-time arrivals, and mishandled baggage rates simultaneously.
- Avoid relying on a single summary metric to represent complex enterprise reliability.

3

Benchmark Discipline 🎯

- Define exact peer comparison sets, seasonal weather adjustments, and schedule boundaries.
- Match performance claims strictly to documented historical quarters and flight networks.



APPENDIX C: TARGET OMNICHANNEL

Reconcile Store and Digital Inventory Data

1

Documented Initiative 🎯

- Target expanded stores-as-hubs fulfillment across its United States retail store network.
- Stores fulfill over 95% of total digital and in-store sales volume nationwide.

2

Ingestion Integrity ⚠️

- Rapid store transactions create timestamp synchronization challenges across enterprise inventory databases.
- Ingestion lags between point-of-sale registers and central databases distort real-time stock levels.

3

Evidence Governance 🛡️

- Audit observation grain, timestamp synchronization, and reconciliation rules before building predictive models.
- Define explicit boundaries separating local store fulfillment nodes from regional distribution warehouses.



Be specific enough that your future self can act on the response.

Individual Exit Ticket

Consolidate your learning before leaving the session

Most important takeaway

State the idea you are most likely to use again.

Question or clarity needed

What still needs clarification?